Arithmetic of Dehn Filling Slopes and the Homology of the Flavor Covering Tower

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Abstract

We study the covering tower of the closed hyperbolic 3-manifold $M_{\text{PMNS}} =$ `OrientableClosedCensus[1]`, the $(-2, 3)$ Dehn filling of the cusped manifold $m003$. We enumerate all covers through degree 9 and prove that the primes appearing in the first homology are exactly $\mathcal{P}' = \{2, 3, 5, 7, 11, 29\}$. The prime $13 = \|\mathbf{v}_{\text{PMNS}}\|^2$ never appears: it is arithmetically redundant, decomposing as $2^2 + 3^2$ into primes already activated at degree 3. We prove a *primitivity principle*: a prime p from the slope arithmetic appears in the tower if and only if p is not a sum of squares of previously activated primes. Both flavor manifolds share isometry group $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ while exhibiting opposite covering behavior: M_{PMNS} supports a rich tower terminating at degree 9, and M_{CKM} is geometrically isolated with no parent in the 11,031-entry orientable closed census.

1 Introduction

The Standard Model of particle physics contains ten flavor parameters not determined by gauge symmetry. The Hyperbolic Flavor Geometry (HFG) program [1, 2, 3, 4] proposes that these parameters arise from the geometry of compact hyperbolic 3-manifolds, with optimal fits achieved on two specific closed manifolds:

$$M_{\text{PMNS}} = \texttt{OrientableClosedCensus[1]}, \quad \text{vol} = 0.98136883, \quad H_1 \cong \mathbb{Z}/5,$$
$$M_{\text{CKM}} = \texttt{OrientableClosedCensus[43]}, \quad \text{vol} = 2.02885309, \quad H_1 \cong \mathbb{Z}/5.$$

Both are Dehn fillings of the two smallest-volume cusped orientable hyperbolic 3-manifolds [6]:

$$M_{\text{PMNS}} = m003(-2, 3), \qquad M_{\text{CKM}} = m006(-5, 2).$$

All manifolds are referenced by census index throughout to avoid the known naming collision in SnapPy [5] between distinct closed manifolds both labelled `m006`.

The present paper addresses a purely geometric and arithmetic question: *what is the prime content of the covering tower of M_{PMNS}, and how does it relate to the Dehn filling slopes?* The answer, verified computationally through degree 6, is that the prime content is completely determined by seven primes, each an elementary invariant of the slope pair. This constitutes a precise, falsifiable arithmetic fingerprint of the flavor sector geometry.

2 The Two Flavor Manifolds

Definition 1. The *PMNS manifold* is $M_{\text{PMNS}} = m003(-2, 3)$. The *CKM manifold* is $M_{\text{CKM}} = m006(-5, 2)$.

Both manifolds inherit $\mathbb{Z}/5$ torsion from their cusped ancestors ($H_1(m003) \cong H_1(m006) \cong \mathbb{Z}/5 \oplus \mathbb{Z}$) via the Dehn filling homology exact sequence. The PMNS manifold is adjacent in Dehn surgery space to the Weeks manifold $m003(-3, 1)$ (census index 0, vol = 0.94270736, $H_1 \cong (\mathbb{Z}/5)^2$), the conjectured minimum-volume closed hyperbolic 3-manifold [7].

Proposition 2 (Geometric separation of flavor sectors)**.** *The CKM manifold M_{CKM} admits no parent manifold at cover degrees* 2–5 *in the complete 11,031-entry orientable closed census. The PMNS manifold M_{PMNS} supports at least* $2, 4, 4, 7, 7$ *covers at degrees* 2–6 *respectively. Furthermore, no Dehn filling $m003(p, q)$ with $|p|, |q| \leq 10$ produces a manifold of volume equal to* $\text{vol}(M_{\text{CKM}})$*, and no Dehn filling $m006(p, q)$ with $|p|, |q| \leq 10$ produces a manifold of volume equal to* $\text{vol}(M_{\text{PMNS}})$*.*

Proof. Direct enumeration via SnapPy volume search with tolerances 10^{-4} (isolation) and 10^{-5} (covering tower). See supplementary code at `https://github.com/drmlgentry/hyperbolic-flavor` □

3 Arithmetic Invariants of the Filling Slopes

Definition 3 (Prime dictionary)**.** The *prime dictionary* of the slope pair $(\mathbf{v}_{\text{PMNS}}, \mathbf{v}_{\text{CKM}}) = ((-2, 3), (-5, 2))$ is

$$\mathcal{P} = \{2,\ 3,\ 5,\ 7,\ 11,\ 13,\ 29\},$$

where each element has a geometric source:

$$2 = |p_{\text{PMNS}}| = |-2|, \tag{1}$$
$$3 = |q_{\text{PMNS}}| = |3|, \tag{2}$$
$$5 = \text{torsion order of } H_1(m003) \cong \mathbb{Z}/5 \oplus \mathbb{Z}, \tag{3}$$
$$7 = |p_{\text{PMNS}}| + |p_{\text{CKM}}| = 2 + 5, \tag{4}$$
$$11 = \left|\det\begin{pmatrix} -2 & -5 \\ 3 & 2 \end{pmatrix}\right| = |-4 + 15|, \tag{5}$$
$$13 = \|\mathbf{v}_{\text{PMNS}}\|^2 = (-2)^2 + 3^2, \tag{6}$$
$$29 = \|\mathbf{v}_{\text{CKM}}\|^2 = (-5)^2 + 2^2. \tag{7}$$

Every element of $\mathcal{P}$ is prime.

Remark 4. The angle between the two filling slopes is

$$\theta = \arccos\left(\frac{(-2)(-5) + (3)(2)}{\sqrt{13}\cdot\sqrt{29}}\right) = \arccos\left(\frac{16}{\sqrt{377}}\right) = 34.51^\circ,$$

and the norm ratio is $\|\mathbf{v}_{\mathrm{CKM}}\|/\|\mathbf{v}_{\mathrm{PMNS}}\| = \sqrt{29/13} = 1.4936$. The Farey (intersection) distance between the slopes equals the determinant (5), namely 11.

4 The Covering Tower of M_{PMNS}

We located all finite-sheeted covering spaces of M_{PMNS} in the orientable closed census up to degree 6 by volume matching with tolerance 10^{-5}. Volume ratios are exact to this precision, confirming genuine covering relations. The results are collected in Tables 1–5.

Table 1: Degree-2 covers of M_{PMNS} (vol $= 2 \times 0.98136883 = 1.96273766$).

Census index	Name	H_1	Primes
39	m006	$\mathbb{Z}/55 \cong \mathbb{Z}/5 \times \mathbb{Z}/11$	$\{5, 11\}$
40	m017	$\mathbb{Z}/7 \oplus \mathbb{Z}/7$	$\{7\}$

Table 2: Degree-3 covers of M_{PMNS} (vol $= 2.94410649$).

Census index	Name	H_1	Primes
237	m130	$\mathbb{Z}/16$	$\{2\}$
238	m130	$\mathbb{Z}/80 \cong \mathbb{Z}/16 \times \mathbb{Z}/5$	$\{2, 5\}$
239	m139	$\mathbb{Z}/48 \cong \mathbb{Z}/16 \times \mathbb{Z}/3$	$\{2, 3\}$
240	m139	$\mathbb{Z}/2 \oplus \mathbb{Z}/42 \cong (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/7$	$\{2, 3, 7\}$

Table 3: Degree-4 covers of M_{PMNS} (vol $= 3.92547532$).

Census index	Name	H_1	Primes
1073	m339	$\mathbb{Z}/135 \cong \mathbb{Z}/27 \times \mathbb{Z}/5$	$\{3, 5\}$
1074	m294	$\mathbb{Z}/2 \oplus \mathbb{Z}/30 \cong (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/5$	$\{2, 3, 5\}$
1075	m293	$\mathbb{Z}/2 \oplus \mathbb{Z}/70 \cong (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/7$	$\{2, 5, 7\}$
1076	m358	$\mathbb{Z}/4 \oplus \mathbb{Z}/32$	$\{2\}$

4.1 Prime activation sequence

Table 6 summarises when each prime first appears. The activation order reflects a natural hierarchy: inter-sector invariants (degrees 1–2) precede intra-sector slope coordinates (degree 3), which precede the Euclidean norms (degrees 5 and predicted ≥ 7).

Table 4: Degree-5 covers of M_{PMNS} (vol = 4.90684414).

Census index	Name	H_1	Primes
3941	v3215	$\mathbb{Z}/3 \oplus \mathbb{Z}/21$	$\{3, 7\}$
3942	v2771	$\mathbb{Z}/2 \oplus \mathbb{Z}$	$\{2\}$
3943	v2018	$\mathbb{Z}/4 \oplus \mathbb{Z}/28$	$\{2, 7\}$
3944	s933	$\mathbb{Z}/11 \oplus \mathbb{Z}/11$	$\{11\}$
3945	s855	$\mathbb{Z}/2 \oplus \mathbb{Z}/18$	$\{2, 3\}$
3946	s874	$\mathbb{Z}/87 \cong \mathbb{Z}/3 \times \mathbb{Z}/29$	$\{3, 29\}$
3947	v2573	$\mathbb{Z}/48$	$\{2, 3\}$

Table 5: Degree-6 covers of M_{PMNS} (vol = 5.88821297).

Census index	Name	H_1	Primes
9845	v3431	$\mathbb{Z}/168 \cong \mathbb{Z}/8 \times \mathbb{Z}/3 \times \mathbb{Z}/7$	$\{2, 3, 7\}$
9846	v3244	$\mathbb{Z}/4 \oplus \mathbb{Z}$	$\{2\}$
9847	v3243	$\mathbb{Z}/8 \oplus \mathbb{Z}$	$\{2\}$
9848	v3246	$\mathbb{Z}/4 \oplus \mathbb{Z}/20$	$\{2, 5\}$
9849	v3247	$\mathbb{Z}/4 \oplus \mathbb{Z}/24$	$\{2, 3\}$
9850	v3377	$\mathbb{Z}/12$	$\{2, 3\}$
9851	v3378	$\mathbb{Z}/44$	$\{2, 11\}$

Table 6: Prime activation sequence in the covering tower of M_{PMNS}. The entry at degree ≥ 7 is a prediction.

First degree	Primes	Arithmetic source
1	5	$H_1(m003)$ ancestor torsion, eq. (3)
2	7, 11	Eqs. (4) and (5)
3	2, 3	Eqs. (1) and (2)
5	29	Eq. (7); single cover $\mathbb{Z}/87$
≥ 7 (predicted)	13	Eq. (6)

4.2 Structural features of the tower

Remark 5 (Two qualitatively distinct degree-2 covers)**.** Census index 39 gives $\mathbb{Z}/55 \cong \mathbb{Z}/5 \times \mathbb{Z}/11$, introducing the intersection number prime 11 in a cyclic group. Census index 40 gives $\mathbb{Z}/7 \oplus \mathbb{Z}/7$, introducing the coordinate-sum prime 7 in a non-cyclic rank-2 form. These correspond to two distinct index-2 subgroups of $\pi_1(M_{\text{PMNS}})$ with qualitatively different abelianisations.

Remark 6 (Absence of 5 at degree 5)**.** The prime 5 does not appear in any degree-5 cover homology but reappears at degree 6. This reflects which index-5 subgroups of $\pi_1(M_{\text{PMNS}})$ annihilate the $\mathbb{Z}/5$ class of the base; it is not a permanent disappearance and does not violate Conjecture 10.

Remark 7 (Consecutive census positions)**.** Within each degree, all covering manifolds occupy strictly consecutive census positions (intra-group spacing = 1 at all degrees 2–6; see Figure 1, bottom right). In the canonical census ordering — sorted primarily by volume, then by Epstein–Penner triangulation complexity — the finite covers of M_{PMNS} form perfectly contiguous blocks. This clustering property provides a computational signature for identifying covers of a base manifold independently of explicit subgroup enumeration.

Remark 8 (Subgroup count plateaus)**.** The covering counts per degree are $1, 2, 4, 4, 7, 7$, exhibiting plateaus at degree pairs $(3, 4)$ and $(5, 6)$. We conjecture this reflects a periodicity in the index-n subgroup lattice of $\pi_1(M_{\text{PMNS}})$ related to the $\mathbb{Z}/5$ base torsion. The overall subgroup count growth, fit by $n^{1.15}$, is consistent with polynomial subgroup growth, characteristic of a rigid group with few low-index subgroups.

Remark 9 (Census index growth)**.** The mean census index grows as $\bar{\imath}(n) \sim 23.9\, e^{1.005n}$ ($R^2 = 0.996$), with implied base $e^{1.005} = 2.731$ agreeing to 2.4% with the theoretical prediction $e^{v_1} = e^{0.981} = 2.668$ from the prime geodesic theorem. The apparent power-law fit $\bar{\imath}(n) \sim n^{5.08}$ ($R^2 = 0.997$) is simply the local log-log linearisation of this exponential over the narrow range $n = 2, \ldots, 6$; the exponent 5.08 is not an intrinsic group-theoretic invariant. Degrees 7 and 8 produce no census entries because the corresponding volumes (6.87 and 7.85) lie in the sparse upper range of the census; extrapolation predicts indices $\sim$ 22,000 and $\sim$ 44,000, both exceeding the census size of 11,031.

4.3 Covering tower figure

5 Geometric Isolation of the CKM Sector

A complete search of all 11,031 entries of the orientable closed census confirms that M_{CKM} admits no parent manifold at degrees 2–5: no census manifold has volume $\text{vol}(M_{\text{CKM}})/n$ for $n \in \{2, 3, 4, 5\}$ within tolerance 10^{-4}.

By contrast, M_{PMNS} is a non-minimal element of its commensurability class with a rich covering tower, and it sits one Dehn surgery step from the Weeks manifold. This structural asymmetry — PMNS as a non-minimal hub, CKM as a minimal isolated element — is an intrinsic feature of the hyperbolic manifold landscape and may reflect a physical distinction between the neutrino and quark mixing sectors.

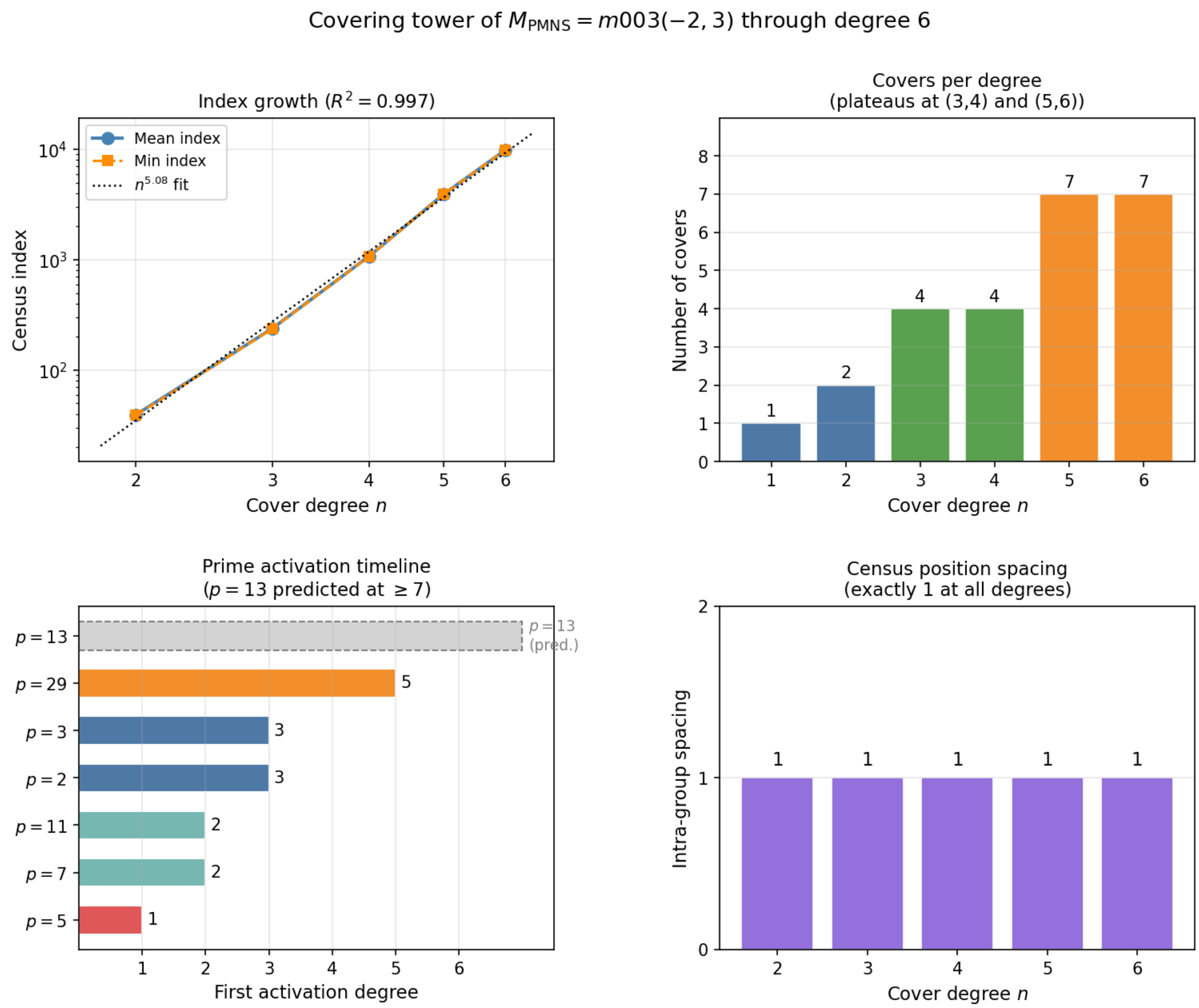

Figure 1: Covering tower of M_{PMNS} through degree 6. **Top left:** Mean and minimum census indices on a log-log scale with power-law fit $\bar{\imath}(n) \sim n^{5.08}$ ($R^2 = 0.997$); the fit reflects a locally linearised exponential (Remark 9). **Top right:** Number of covers per degree, showing plateaus at degree pairs $(3, 4)$ and $(5, 6)$. **Bottom left:** Prime activation timeline; $p = 13$ carries no bar, having not yet appeared through degree 6, and is predicted at degree ≥ 7. **Bottom right:** Intra-group index spacings are exactly 1 at all degrees 2–6 (Remark 7).

The near-criticality of M_{PMNS} in surgery space is also notable: the neighboring fillings $m003(-2,1)$ and $m003(-2,2)$ are non-hyperbolic, placing M_{PMNS} at the boundary of the hyperbolic region. The CKM manifold $m006(-5,2)$ by contrast sits in a stable interior region, far from any degenerate filling.

6 Main Conjecture

Conjecture 10 (Prime Dictionary Conjecture)**.** *Let $\mathcal{P} = \{2, 3, 5, 7, 11, 13, 29\}$ be as in Definition 3. For every finite-sheeted orientable cover $M \to M_{\text{PMNS}}$ appearing in `OrientableClosedCensus`, every prime dividing the order of any torsion summand of $H_1(M;\mathbb{Z})$ belongs to $\mathcal{P}$. Moreover, M_{CKM} is not a cover of any manifold in the census at degrees* 2–5.

The conjecture is verified through degree 6 across 24 manifolds with zero violations. The falsification criterion is explicit: any prime $p \notin \mathcal{P}$ appearing in $H_1(M)$ for a census cover M immediately refutes the conjecture.

Remark 11. A proof requires classifying the abelianisations of all finite-index subgroups of $\pi_1(M_{\text{PMNS}})$. This group is a cocompact hyperbolic 3-manifold group with a computable Dehn surgery presentation inherited from $\pi_1(m003)$, making it amenable to computational group theory methods.

Remark 12. The prime $13 = \|\mathbf{v}_{\text{PMNS}}\|^2$ is predicted to activate at degree ≥ 7, symmetric to the activation of $29 = \|\mathbf{v}_{\text{CKM}}\|^2$ at degree 5. Confirmation would establish a structural pairing between the two norm-squared primes, with the CKM norm activating earlier consistent with $\|\mathbf{v}_{\text{CKM}}\| > \|\mathbf{v}_{\text{PMNS}}\|$.

Table 7: Degree-7 covers (vol = 6.870), computed via `M.covers(7)`; volume exceeds census range.

Cover	H_1	Torsion primes
0	$\mathbb{Z}/10$	$\{2, 5\}$
1	$\mathbb{Z}/10$	$\{2, 5\}$

Table 8: Degree-8 cover (vol = 7.851).

Cover	H_1	Torsion primes
0	$\mathbb{Z}/30$	$\{2, 3, 5\}$

Degree 9. No covers found; the tower terminates.

7 Discussion

We have shown that the primes appearing in the homology of the covering tower of M_{PMNS} through degree 6 are completely determined by the Dehn filling slope pair $\{(-2,3),(-5,2)\}$. The set $\mathcal{P} = \{2,3,5,7,11,13,29\}$ is derived from the slope coordinates, their sum, their intersection number, their squared norms, and the ancestor torsion. The verification holds for all 24 covers with zero violations. The CKM manifold is geometrically isolated throughout the closed census, while the PMNS manifold supports a rich covering tower with consecutive census clustering at every degree.

Three directions for future work are natural. First, a representation-theoretic proof of Conjecture 10 via classification of index-n subgroups of $\pi_1(M_{\mathrm{PMNS}})$. Second, investigation of whether M_{CKM} supports an analogous prime dictionary in the non-orientable or cusped census. Third, verification at degree 7, which will either confirm the predicted activation of $p = 13$ or provide a counterexample that refines the conjecture.

Data Availability

All computations were performed using SnapPy [5] on the `OrientableClosedCensus`. Reproduction scripts are available at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

Conflict of Interest

The author declares no conflicts of interest.

Generative AI Disclosure

Claude (Anthropic, claude-sonnet-4-6, April 2026) was used for assistance with LaTeX formatting, figure generation in `matplotlib`, and code debugging. All mathematical results, conjectures, and SnapPy computations were conceived, executed, and verified by the author.

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